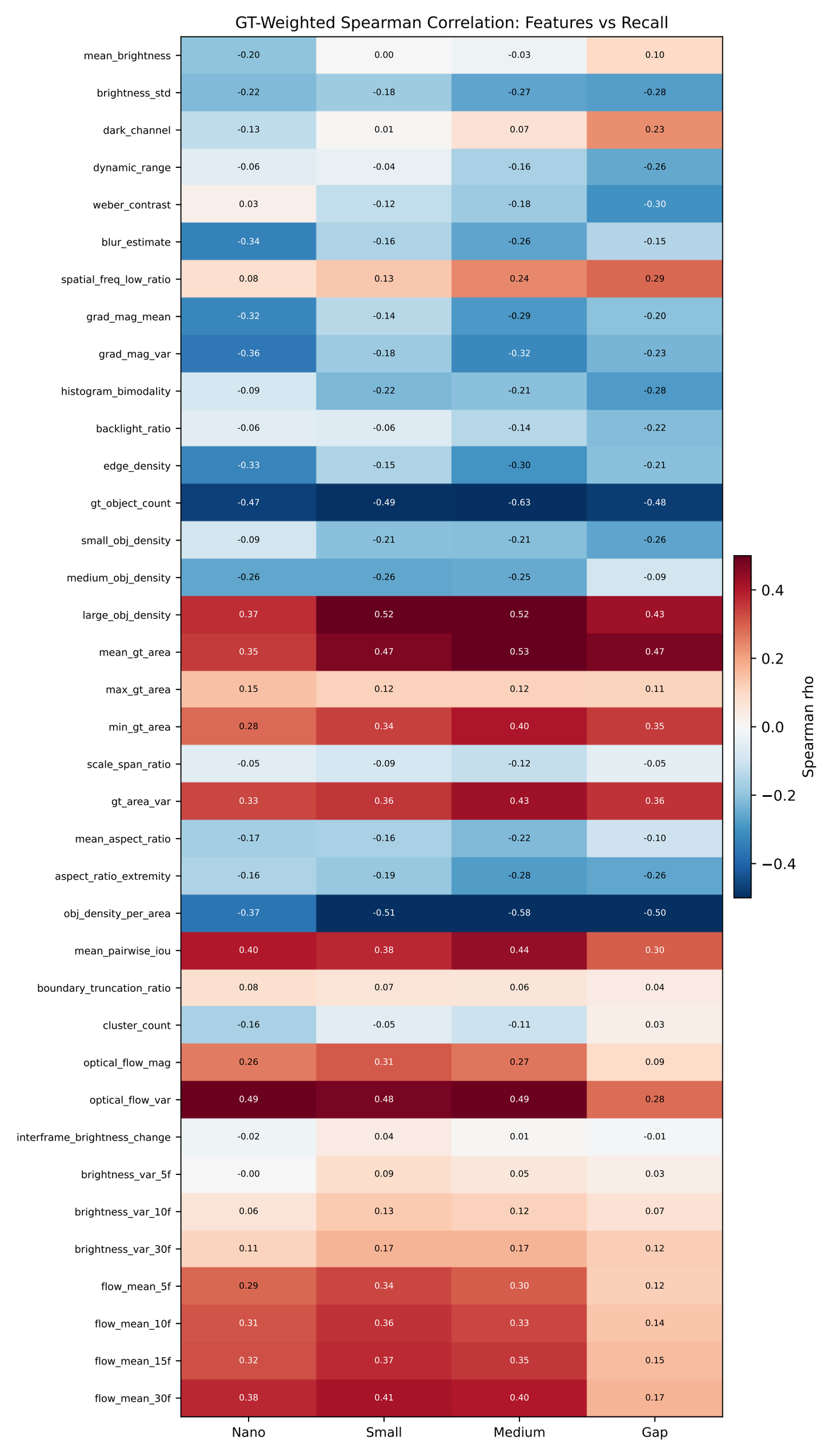
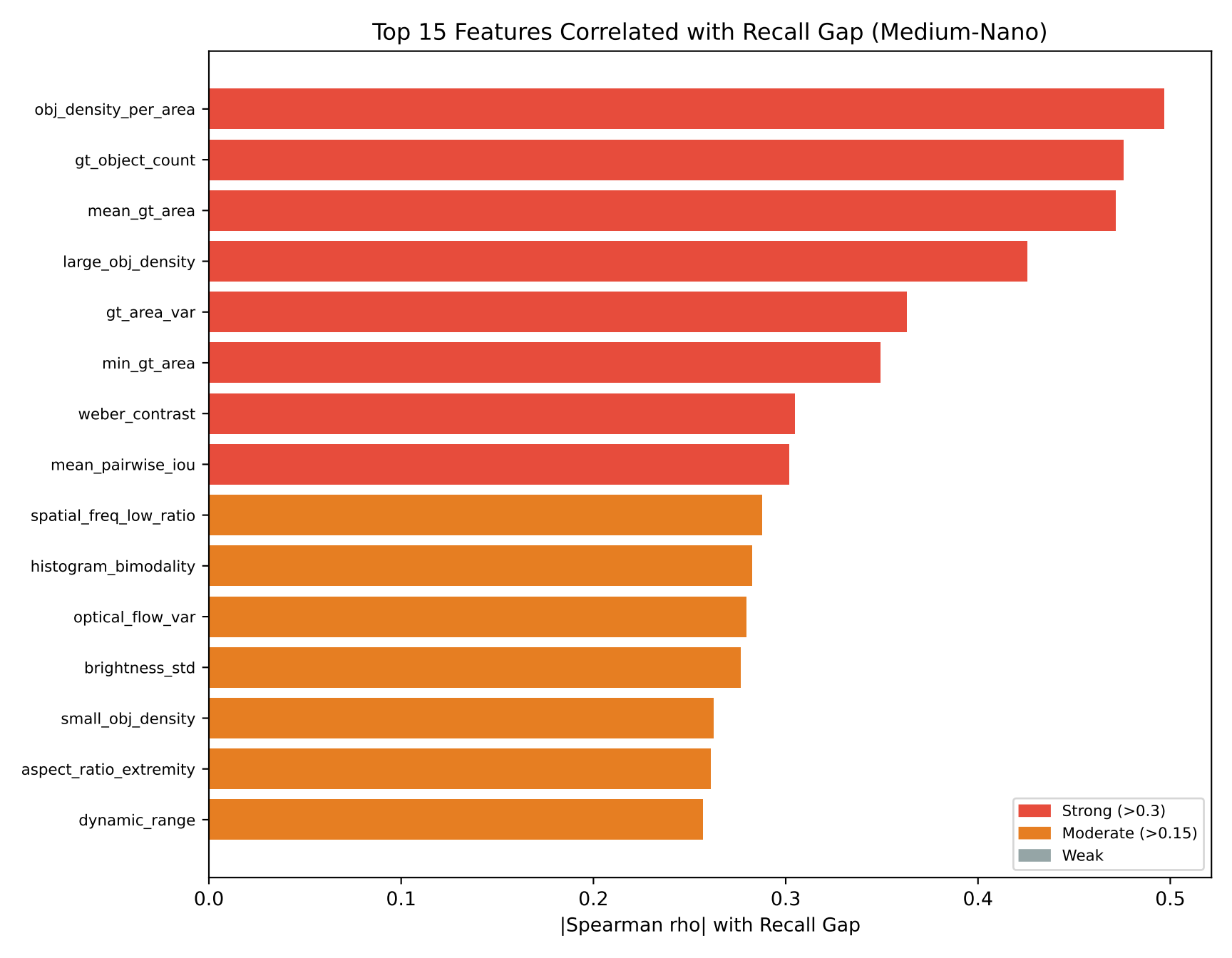
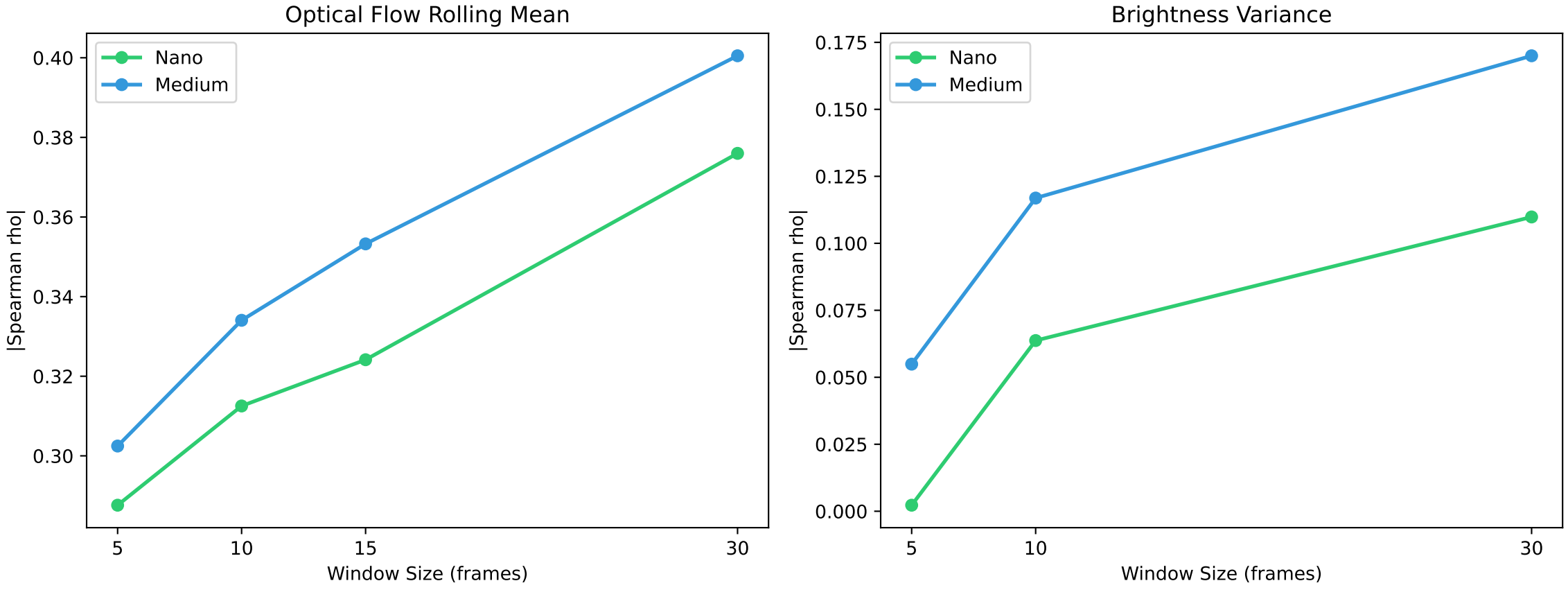

GT-Weighted Spearman Correlation: Features vs Recall
mean_brightness -0.20 0.00 -0.03 0.10
brightness_std -0.22 -0.18 -0.27 -0.28
dark_channel -0.13 0.01 0.07 0.23
dynamic_range -0.06 -0.04 -0.16 -0.26
weber_contrast 0.03 -0.12 -0.18 -0.30
blur_estimate -0.34 -0.16 -0.26 -0.15
spatial_freq_low_ratio 0.08 0.13 0.24 0.29
grad_mag_mean -0.32 -0.14 -0.29 -0.20
grad_mag_var -0.36 -0.18 -0.32 -0.23
histogram_bimodality -0.09 -0.22 -0.21 -0.28
backlight_ratio -0.06 -0.06 -0.14 -0.22
edge_density -0.33 -0.15 -0.30 -0.21
gt_object_count -0.47 -0.49 -0.63 -0.48
small_obj_density -0.09 -0.21 -0.21 -0.26
medium_obj_density -0.26 -0.26 -0.25 -0.09
large_obj_density 0.37 0.52 0.52 0.43
mean_gt_area 0.35 0.47 0.53 0.47
max_gt_area 0.15 0.12 0.12 0.11
min_gt_area 0.28 0.34 0.40 0.35
scale_span_ratio -0.05 -0.09 -0.12 -0.05
gt_area_var 0.33 0.36 0.43 0.36
mean_aspect_ratio -0.17 -0.16 -0.22 -0.10
aspect_ratio_extremity -0.16 -0.19 -0.28 -0.26
obj_density_per_area -0.37 -0.51 -0.58 -0.50
mean_pairwise_iou 0.40 0.38 0.44 0.30
boundary_truncation_ratio 0.08 0.07 0.06 0.04
cluster_count -0.16 -0.05 -0.11 0.03
optical_flow_mag 0.26 0.31 0.27 0.09
optical_flow_var 0.49 0.48 0.49 0.28
interframe_brightness_change -0.02 0.04 0.01 -0.01
brightness_var_5f -0.00 0.09 0.05 0.03
brightness_var_10f 0.06 0.13 0.12 0.07
brightness_var_30f 0.11 0.17 0.17 0.12
flow_mean_5f 0.29 0.34 0.30 0.12
flow_mean_10f 0.31 0.36 0.33 0.14
flow_mean_15f 0.32 0.37 0.35 0.15
flow_mean_30f 0.38 0.41 0.40 0.17
Nano
Small
Medium
Gap
Spearman rho
0.4
0.2
0.0
−0.2
−0.4

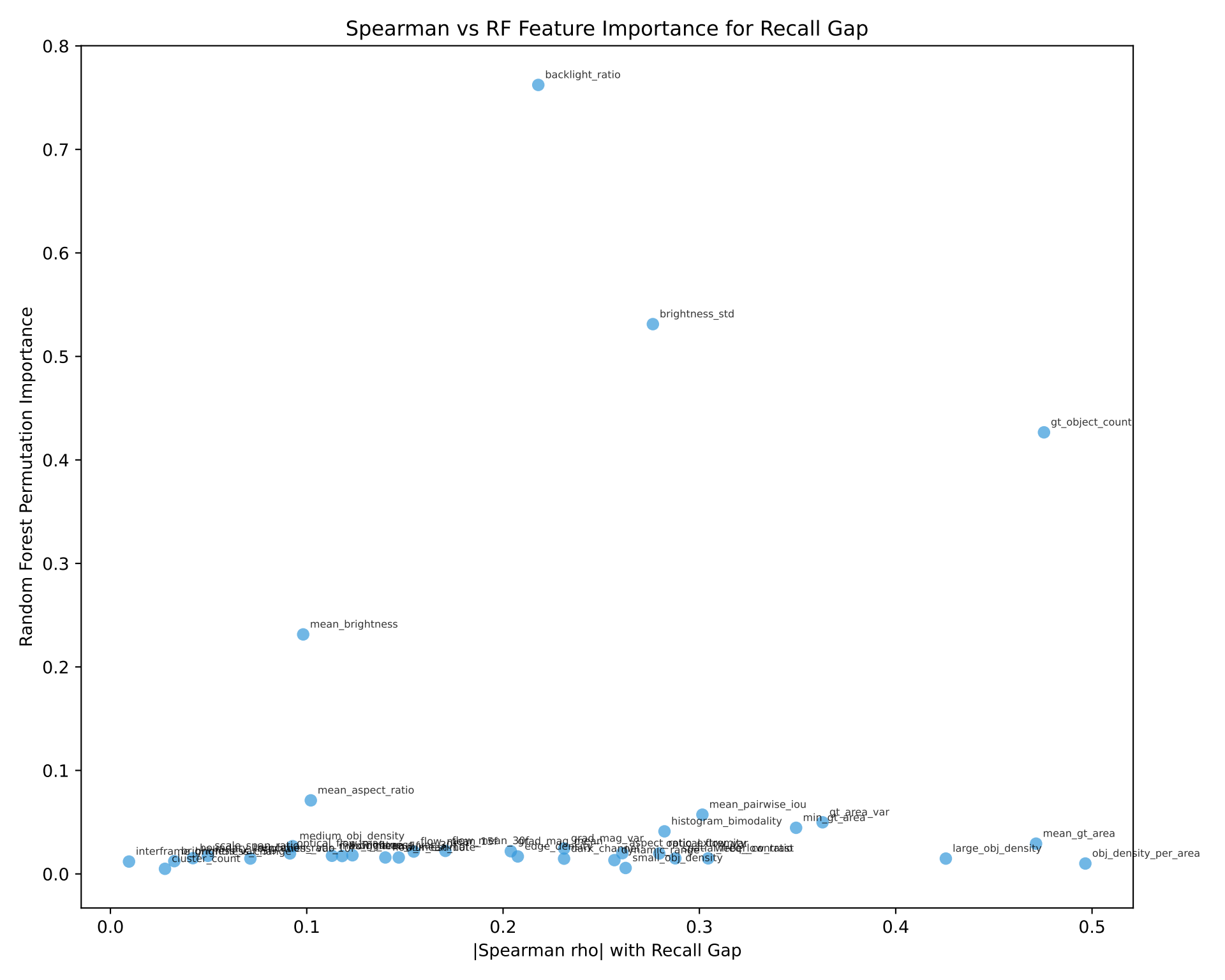

Top 15 Features Correlated with Recall Gap (Medium-Nano)
obj_density_per_area
gt_object_count
mean_gt_area
large_obj_density
gt_area_var
min_gt_area
weber_contrast
mean_pairwise_iou
spatial_freq_low_ratio
histogram_bimodality
optical_flow_var
brightness_std
small_obj_density
aspect_ratio_extremity
dynamic_range
Strong (>0.3)
Moderate (>0.15)
Weak
0.0
0.1
0.2
0.3
0.4
0.5
|Spearman rho| with Recall Gap


Temporal Horizon Analysis: Correlation Strength vs Window Size
Optical Flow Rolling Mean
Brightness Variance
Nano
Medium
|Spearman rho|
Window Size (frames)
0.40
0.38
0.36
0.34
0.32
0.30
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
5
10
15
30

Feature Pairwise Spearman Correlation (x = |rho|>0.85)

mean_brightness
brightness_std
dark_channel
dynamic_range
weber_contrast
blur_estimate
spatial_freq_low_ratio
grad_mag_mean
grad_mag_var
histogram_bimodality
backlight_ratio
edge_density
gt_object_count
small_obj_density
medium_obj_density
large_obj_density
mean_gt_area
max_gt_area
min_gt_area
scale_span_ratio
gt_area_var
mean_aspect_ratio
aspect_ratio_extremity
obj_density_per_area
mean_pairwise_iou
boundary_truncation_ratio
cluster_count
optical_flow_mag
optical_flow_var
interframe_brightness_change
brightness_var_5f
brightness_var_10f
brightness_var_30f
flow_mean_5f
flow_mean_10f
flow_mean_15f
flow_mean_30f

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Spearman vs RF Feature Importance for Recall Gap
Random Forest Permutation Importance
|Spearman rho| with Recall Gap
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
backlight_ratio
brightness_std
gt_object_count
mean_brightness
mean_aspect_ratio
mean_pairwise_iou
histogram_bimodality
gt_area_var
min_gt_area
mean_gt_area
large_obj_density
obj_density_per_area
medium_obj_density
cluster_count
small_obj_density